

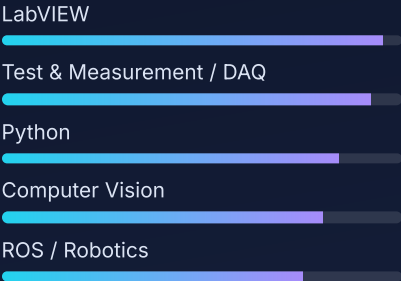
KIRAN SHIVAKUMAR

Application Engineer · LabVIEW · Robotics

CONTACT

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CORE SKILLS



AI & VISION

- OpenCV
- TensorFlow
- MediaPipe
- YOLO v5/v8/v11/26
- DeepSORT

ROBOTICS PLATFORM

- ROS
- Gazebo
- RViz
- ROS SLAM

EDUCATION

B.E. in Mechatronics
The Oxford College of Engineering
2017 – 2021 ·
8.1 CGPA

PROFESSIONAL PROFILE

Dedicated Application Engineer with **5+ years** of experience designing, developing, and deploying **LabVIEW-based Automated Test Equipment (ATE)** and Test & Measurement systems for aerospace, space, nuclear, and power industries. Experienced with instrument control over SCPI/GPIB, multi-sensor data acquisition, and integrating **Python, ROS, and Computer Vision** for intelligent robotics and predictive monitoring.



WORK EXPERIENCE

Application Engineer Feb 2024 – Present
Ducom Aerospace Pvt Ltd · Bengaluru, India

- Develop **LabVIEW-based software** for Test and Measurement related to Tribology testing systems.
- Capture and analyze sensor data: **force, temperature, humidity, vibration, and acoustic emission**.
- Build Computer Vision pipelines for image processing, annotation, segmentation, facial detection, and tracking human movement for surveillance using **YOLO, OpenCV, and DeepSORT**.
- Develop robotic mapping and live video visualization for a land platform using **ROS, Gazebo, and SLAM**.

Tech: LabVIEW, Python, OpenCV, TensorFlow, MediaPipe, YOLO, DeepSORT, ROS, Gazebo, RViz, SLAM

LabVIEW Developer Sep 2021 – Jan 2024
Access Automation Pvt Ltd · Bengaluru, India

- Architected and delivered **7+ automated test & DAQ systems** for premier clients including **ISRO (SAC), NPCIL, Tata Power, LG India, GACL, Electrohms, and Oman Fiber Optics**.
- Interfaced laboratory instrumentation from Keysight, Yokogawa, HIOKI, Anritsu, and EXFO with Mitsubishi FX5U PLCs.
- Developed automated test sequences, remote monitoring dashboards, and compliance report generators.
- Delivered continuous optimization recommendations and engineered product innovations.

Tech: LabVIEW, ATE, DAQ, SCPI, GPIB, Modbus, RS-485, PLC, Report Generation

FEATURED CLIENT PROJECTS

DAQ & Reporting Software for Antenna Testing SAC · ISRO
Configures antenna testing parameters (frequency, bandwidth, measurement specs), controls DUT/feed positioners, and measures radiation patterns.
HW: Keysight MXG N5183B, PNA-X N5264B, Power Meter, Spectrum Analyzer E4446A, Position Controllers

Transformer Testing Bench and Software TATA POWER
Automated test bench executing no-load, full-load, temperature rise, vector group, DVDF, high-voltage, turn-ratio, and insulation resistance tests.
HW: Yokogawa WT500, Motwani XTRM-3A, Motwani XWRM-25A, Motwani 5kV IR Tester, FX5U PLC

KIRAN SHIVAKUMAR

INSTRUMENTS INTEGRATED

- Keysight PNA-X N5264B
- Keysight MXG N5183B
- Keysight 33210A AWG
- Keysight DSO-X 2012A
- Spectrum Analyzer E4446A
- Yokogawa WT500 Analyzer
- Yokogawa GM10 (19 Units)
- HIOKI PW3335 Power Meter
- Anritsu MT9085 OTDR
- EXFO FTB-500 OTDR
- Advantech ADAM 4050/4060
- Motwani Turn-Ratio XTRM-3A
- Motwani Winding Res. XWRM-25A
- Motwani 5kV Diagnostic Tester
- Mitsubishi FX5U PLC

COMMUNICATION PROTOCOLS

- SCPI Commands
- GPIO Interface
- RS-485 Serial
- RS-232 Serial
- Modbus Protocol
- CAN Bus
- HART Protocol
- TCP/IP Socket
- Telnet
- FTP
- USB DAQ

INDUSTRY DOMAINS

- Aerospace & Defense
- Space & RF Testing
- Nuclear Safety Monitoring
- Power & Transformers
- Fiber Optics Telecom
- Consumer Appliances
- Tribology DAQ
- Autonomous Robotics

LANGUAGES

- English
- Hindi
- Kannada
- Tamil

OPERATING SYSTEMS

- Windows
- Linux (Ubuntu)

FEATURED CLIENT PROJECTS (CONTD.)

Perimeter Automated Detection System

NPCIL · KAKRAPAR ATOMIC POWER

Monitors and acknowledges safety alarms triggered by microwave detectors and PIR perimeter sensors across nuclear plant zones.

HW: Advantech ADAM-4050, Advantech ADAM-4060 Modules

Power Consumption Test Automated Software

LG INDIA

Automated test sequence acquiring electrical parameters from power analyzers and proximity sensors with automatic pass/fail reporting.

HW: HIOKI PW3335 Power Meter, Mitsubishi FX5U PLC, Proximity Sensors

Cell Voltage Monitoring System (19 Networked Units)

GACL INDIA

Networked DAQ architecture interfacing 19 Yokogawa GM10 units across a shared industrial LAN for continuous remote cell monitoring.

HW: Yokogawa GM10 Data Acquisition System (19 Nodes), TCP/IP

ATE Software for Frequency Response

ELECTROHMS INDIA

Automated control of arbitrary waveform generator and oscilloscope to sweep frequencies, compute RMS values, and plot response in dB.

HW: Keysight 33210A AWG, Keysight DSO-X 2012A Oscilloscope

ConnectOTDR Fiber Optic Testing System

OMAN FIBER OPTICS

Multi-instrument optical testing suite evaluating wavelength, resolution, pulse width, and fault distance across fiber runs.

HW: EXFO FTB-500 Platform, Anritsu MT9085 OTDR

FEATURED OPEN-SOURCE PROJECTS

LabVIEW ↔ Python AI Bridge

Hybrid architecture streaming LabVIEW DAQ sensor arrays into Python ML for real-time predictive anomaly detection.

github.com/kiran10299/LabVIEW-Python-AI-Bridge

AI Hand-Gesture PC Controller

Computer Vision desktop app controlling mouse cursor, clicking, and air-scrolling via MediaPipe and OpenCV HUD.

github.com/kiran10299/AI-Hand-Gesture-Controller

Desktop Super Tools

System productivity utilities featuring a background watchdog file sorter and global hotkey translucent dropdown notepad.

github.com/kiran10299/Desktop-Super-Tools

Straw-Hat Reminders

Procedural vector animated desktop companion running in system tray with active screen-time health monitoring.

github.com/kiran10299/Straw-Hat-Reminders